

DESIGN AND ANALYSIS OF ALGORITHMS

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Experiment 11: Heap Sort

Code:

```
def heapify(arr, n, i):
    largest = i
    left = 2 * i + 1
    right = 2 * i + 2
    if left < n and arr[left] > arr[largest]:
        largest = left
    if right < n and arr[right] > arr[largest]:
        largest = right
    if largest != i:
        arr[i], arr[largest] = arr[largest], arr[i]
        heapify(arr, n, largest)

def heap_sort(arr):
    n = len(arr)
    for i in range(n // 2 - 1, -1, -1):
        heapify(arr, n, i)
    for i in range(n - 1, 0, -1):
        arr[0], arr[i] = arr[i], arr[0]
        heapify(arr, i, 0)

arr = [4, 10, 3, 5, 1]
heap_sort(arr)
print("Sorted Array:", arr)
```

Output:

```
===== RESTART: C:/Users/hk  
Sorted Array: [1, 3, 4, 5, 10]
```